



Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges

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ABSTRACT

This paper investigates the optimal design of the Maxwell tuned mass-damper-inerter (M-TMDI) for mitigating the vortex-induced vibration (VIV) in bridges. The M-TMDI consists of a three-element tuned mass damper (TMD) and an inerter. Considering that the bridge deck is a multiple-degree-of-freedom (DOF) system, the inerter location is considered as a design variable of the M-TMDI in our study. The optimal parameters of a specific M-TMDI, in which the end of the inerter is connected to the fixed ground, are analytically given based on a two-DOF system. Furthermore, the optimal parameters of the M-TMDI with any inerter location on the bridge deck are developed in closed-form based on a multiple-DOF system. Finally, numerical analysis on a continuous steel box-girder bridge subjected to the VIV is performed to confirm the optimal design and superiority of the M-TMDI control. The result demonstrates that the optimally designed M-TMDI outperforms the TMD and three-element TMD in the transient amplitude mitigation, steady-state amplitude mitigation, stroke limitation, and static stretching reduction. The optimal control effect of the M-TMDI greatly depends on the defined effective mass ratio, which is function of the inerter location, mode shape, physical mass, and inertial mass.

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1. Introduction

The dynamic vibration absorber or tuned mass damper (TMD) as one of the most reliable dampers is widely used to reduce structural vibrations [1–5,45]. A conventional TMD can absorb and dissipate large amounts of vibration energy from the primary structure through the optimal tuning. However, the TMD control is sensitive to the mistuning phenomenon, even when robust optimally designed [2,5,6]. Semi-active TMDs can adjust the TMD characteristics in real time to weaken the mistuning phenomenon [7]. Since the introduction of semi-active devices, there are some concerns on the controller design compared to the passive control, such as the noise disturbance, state estimation, time delay [8]. Active TMDs have an excellent control effect but also have a high control cost and stability concerns from control algorithms and actuator [9–11]. As a result, the development of passive TMDs is still a popular task in the structural vibration control. These passive TMDs include the nonlinear energy sinks [12,13], viscoelastic TMDs [5,14,15], TMD with negative stiffness [16], partial TMD [17], tuned inerter-based dampers [18,19], and so on.

Among these passive TMDs, the tuned mass-damper-inerter (TMDI) taking advantage of the mass amplification effect from the inerter has attracted the attention in recent years [19–22]. Giaralis et al. [20–22] found that the TMDI has better control effect, stroke limitation, and high-mode damping effect than the TMD in both the seismically excited structure

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and wind-excited tall building. Only a few studies investigated the wind-induced vibration mitigation of bridges using the TMDI. Since the long-span bridge is flexible, the natural frequency of the TMD used in long-span bridges is always smaller than 1 Hz. According to $L = g / (4\pi^2 f_t^2)$, where g is the gravitational acceleration, f_t is the natural frequency of the TMD, its static stretching L due to gravity can reach 1 m when f_t is 0.5 Hz. Besides, the mass ratio of the TMD used in bridges is very small, so that its stroke limitation will become challenging considering the motion space and tuning mechanism. Thanks to the inerter with the characteristics of negative stiffness [23], the TMDI can overcome the two major concerns of mass dampers in flexible bridges [24,25]. Xu et al. [24] and Dai et al. [25] used the TMDI to mitigate the vortex-induced vibration (VIV) in bridges. They found that the TMDI configuration has much smaller static stretching and stroke than the TMD.

Besides, Asami and Nishihara [26,27] proposed a novel three-element TMD, in which the dashpot element is replaced by the Maxwell element, to produce better performance than the TMD. The optimal parameters for the undamped and damped structures are given analytically using the fixed-point method [26], H_2 optimization [27], equivalent linearization method [28], and generalized optimality criteria approach [29]. The comparison of frequency response curves shows that the two-peak reduction using the three-element TMD is not significant compared to that using the TMD [26–29], but the original resonance reduction is significant, as shown in Fig. 2. Based on this feature, it seems that the three-element TMD is more suitable for mitigating the self-excited vibration where the original resonance needs to be controlled than the TMD. For the bridge deck with separated flows, the vortices are shed periodically with a resulting wake that periodically impacts the deck to generate a cross-wind force, resulting in the VIV [30,31]. When the vortex-shedding frequency approaches a natural frequency of the bridge, a large amplitude oscillator called vortex-induced resonance occurs within a certain range of wind velocities [30–32]. Experimental and numerical results showed that the vortex-induced resonance in bridges can be simplified as a van der Pol oscillator with the self-excited and self-limited characteristics [30–33]. In view of the VIV characteristics, it seems that the three-element TMD could have better control effect than the TMD in the VIV mitigation of bridges.

Although some researchers proposed a novel TMD consisting the three-element TMD and inerter, they did not use the damper to mitigate the wind-induced vibration in bridges. Wang et al. [34] developed the closed-form optimization solution for the two types of three-element TMDIs by using the fixed-point method. Javidialesaadi and Wierschem [35] adopted the H_2 and H_∞ optimizations to obtain the closed-form optimal parameters of the three-element TMDI. In their three-element TMDIs [34,35], the end of the inerter is connected to the structure or fixed ground, and the structure is investigated by a single-degree-of-freedom (DOF) system. However, it is difficult to realize the inerter-ground configuration if such mass dampers are installed on the bridge deck, and the inerter-structure configuration has worse control effect than the TMD [24,25,34]. Thus, the current optimization solutions are not applicable to the three-element TMDI used in bridges. In view of the features offered by the three-element TMD and inerter, this paper proposes a generalized three-element TMDI named Maxwell TMDI (M-TMDI) to mitigate the VIV in bridges. The bridge is investigated by a multiple-DOF system, and the inerter location is considered as a variable. In this case, the optimization solution is the generalized solution of the M-TMDI that realizes the static stretching reduction, stroke limitation, and better control effect simultaneously.

This paper is organized as follows. Section 2 discuss the three-element TMD and the inerter-ground M-TMDI, and analytically gives the optimal parameters of the inerter-ground M-TMDI. Section 3 develops a closed-form optimization solution for the M-TMDI with any inerter location on the bridge deck. Section 4 presents a numerical example of a continuous steel box-girder bridge subjected to the VIV to confirm the optimal design and superiority of the M-TMDI control. Section 5 presents the conclusion.

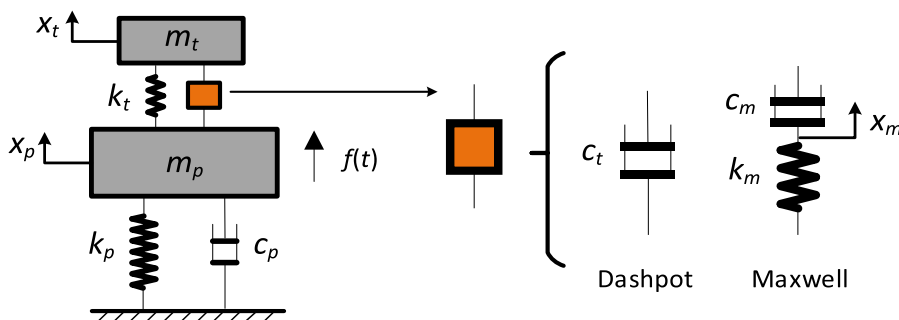


Fig. 1. A representative two-DOF system.

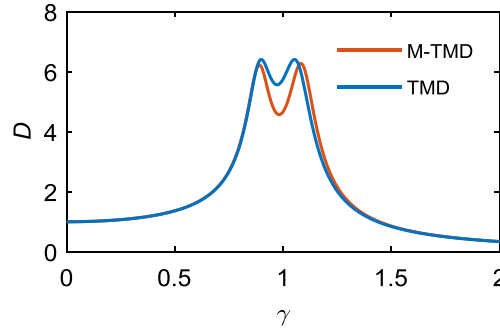


Fig. 2. TMD control and M-TMD control under the harmonic force excitation.

2. Background

2.1. Maxwell tuned mass damper

The primary structure equipped with a mass damper can be investigated by a representative two-DOF system, shown in Fig. 1. The equations of motion of the two-DOF system are written as

$$m_p \ddot{x}_p(t) + c_p \dot{x}_p(t) + k_p x_p(t) = f(t) + u(t) \quad (1)$$

$$m_t \ddot{x}_t(t) + u(t) = 0 \quad (2)$$

where x_p and x_t are the displacements of the primary structure and auxiliary mass relative to the ground, respectively; m_p , c_p and k_p are the mass, damping, and stiffness of the primary structure, respectively; m_t is the mass of the auxiliary mass; f is the external force acting on the primary structure. To give a general form of the two-DOF system, the total force provided by the mass damper u is introduced into Eqs. (1) and (2). When the mass damper is a conventional TMD, u is equal to

$$u(t) = k_t [x_t(t) - x_p(t)] + c_t [\dot{x}_t(t) - \dot{x}_p(t)] \quad (3)$$

where c_t and k_t are the damping and stiffness of the TMD. When the mass damper is a three-element TMD, i.e., M-TMD, in which the dashpot element is replaced by a Maxwell element, shown in Fig. 1, u is equal to

$$u(t) = k_t [x_t(t) - x_p(t)] + c_m [\dot{x}_t(t) - \dot{x}_m(t)] \quad (4)$$

$$c_m [\dot{x}_t(t) - \dot{x}_m(t)] = k_m [x_m(t) - x_p(t)] \quad (5)$$

where x_m is the inner displacement of the Maxwell element relative to the ground; c_m and k_m are the damping and stiffness of the Maxwell element.

If the primary structure is a flexible bridge, the damping ratio of the primary structure can be neglected, because the damping ratio of flexible bridges is always less than 1% [30,31] and the effect of small structural damping ratio on the M-TMD control are very small [27]. Taking Fourier transform of Eqs. (1), (2), (4) and (5), we obtain the displacement dynamic amplification factor (DAF) of the primary structure equipped with a M-TMD [27],

$$D^2 = \frac{|x_p|_{x_0}^2}{\kappa^2 \beta^2 [\beta^2 - (1 + \beta^2 + \mu \beta^2) \gamma^2 + \gamma^4]^2 + 4 \xi_t^2 \gamma^2 [\beta^2 + \kappa \beta^2 - (1 + (1 + \mu)(1 + \kappa) \beta^2) \gamma^2 + \gamma^4]^2} \quad (5)$$

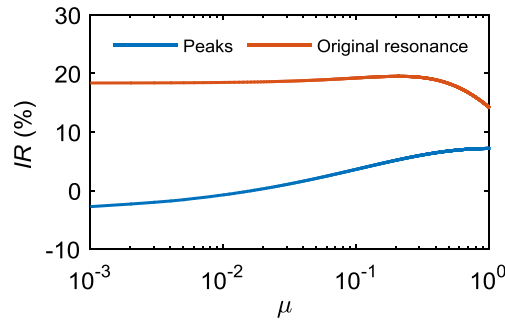
where D is the displacement DAF of the primary structure; $x_0 = f/k_p$ is the static displacement; $\gamma = \omega/\omega_p$, $\beta = \omega_t/\omega_p$, and $\kappa = k_m/k_t$ are the dimensionless parameters; $\mu = m_t/m_p$ is the mass ratio; ω is the excitation frequency; $\omega_p = \sqrt{k_p/m_p}$ is the natural frequency of the primary structure; $\omega_t = \sqrt{k_t/m_t}$, $\xi_t = c_m/(2m_t\omega_t)$. Amasi and Nishihara [26,27] used the H-norm technique to obtain the analytical optimization solutions of the M-TMD. Table 1 presents the expressions of the formulas in different excitations.

Fig. 2 shows the comparison between the TMD and M-TMD controls under the harmonic force excitation. The mass ratio μ is assumed as 5%. The TMD is optimally designed according to Den Hartog's formulas. The maximum displacement DAFs using the TMD and M-TMD are 6.408 and 6.276, respectively, indicating that the superiority of the M-TMD in the peak response reduction is not significant. However, the displacement DAFs between the two peaks using the M-TMD are much smaller than those using the TMD, with a reduction of 18.85% at the original resonance with $\gamma = 1$. Fig. 3 shows the improvement ratio of the M-TMD with varying mass ratios. The improvement ratio here is defined as $IR = (D_T - D_M)/D_T$, where D_T and D_M are the displacement DAFs using TMD and M-TMD controls at two peaks or original resonance. It can be seen that the improvement ratio at two peaks increases with the mass ratio, but the maximum IR is only 7.2% when $\mu = 1$. The improve-

Table 1

Design formulas for the Maxwell TMD [26,27].

Excitation	Index	β	κ	ξ_t
$f_o e^{i\omega t}$	$D = \left \frac{x_p}{f_o} \right $	$\sqrt{\frac{1}{1+\mu} \left(1 - \sqrt{\frac{\mu}{1+\mu}} \right)}$	$2\mu + 2\sqrt{\mu(1+\mu)}$	$\sqrt{\frac{1+r}{r} \cdot \frac{-b - \sqrt{b^2 - 4ac}}{a}}$
	$I_1 = \frac{\sigma_{ss}^2 k_p^2}{2\pi S_0 \omega_p^2}$	$\sqrt{1 - \sqrt{\frac{\mu}{1+\mu}}}$	$2\mu + 2\sqrt{\mu(1+\mu)}$	$\sqrt{\mu + \mu^2 - \frac{3+2\mu}{2} \sqrt{\frac{\mu}{1+\mu}}}$
$\ddot{x}_g e^{i\omega t}$	$I_2 = \frac{\sigma_{ss}^2 \omega_p^3}{2\pi S_0}$	$\sqrt{\frac{1-\sqrt{\mu}}{1+\mu}}$	$\frac{2\sqrt{\mu}}{1-\sqrt{\mu}}$	$(1+\mu) \sqrt{\frac{(1+\mu)(2-\sqrt{\mu})}{2\mu}}$
$r = \sqrt{\frac{1+\mu}{\mu}}, a = -2 - 2r + 5r^2 + 4r^3 - 2r^5 + r^6, b = 2 - 3r^2 - r^4, c = -2 + 2r + r^2$				

**Fig. 3.** The improvement ratio of the M-TMD control with varying mass ratios.

ment ratio at the original resonance is considerable within all the mass ratios, for example, $IR = 19.5\%$ when $\mu = 20\%$. The comparison result indicates that the M-TMD outperforms the TMD in suppressing the structural vibration whose dominant frequency is near the original resonance frequency, such as the VIV in bridges.

To understand the superiority of the M-TMD, the total force provided by the Maxwell element in parallel with a spring can be written as the form of Kelvin element in the frequency domain

$$u(\omega) = \left(\frac{\omega^2 c_m^2 k_m}{k_m^2 + \omega^2 c_m^2} + k_t \right) \Delta + \frac{c_m k_m^2}{k_m^2 + \omega^2 c_m^2} i\omega \Delta \quad (6)$$

where $\Delta = x_t - x_p$ is the displacement of the M-TMD relative to the primary structure.

Substituting dimensionless parameters into Eq. (6) yields

$$\frac{u(\omega)}{m_p \omega_p^2} = \mu \left[\left(\frac{4\kappa \xi_t^2 \beta^2 \gamma^2}{\kappa^2 \beta^2 + 4\xi_t^2 \gamma^2} + \beta^2 \right) \Delta + \frac{2\kappa^2 \xi_t \beta^3}{\omega_p (\kappa^2 \beta^2 + 4\xi_t^2 \gamma^2)} i\omega \Delta \right] = \mu \left(\bar{\beta}^2 \Delta + \frac{2\bar{\xi}_t \bar{\beta}}{\omega_p} i\omega \Delta \right) \quad (7)$$

$$\bar{\beta} = \beta \sqrt{1 + \frac{4\kappa \xi_t^2 \gamma^2}{\kappa^2 \beta^2 + 4\xi_t^2 \gamma^2}} \quad (8)$$

$$\bar{\xi}_t = \frac{\kappa^2 \xi_t \beta^3}{\beta (\kappa^2 \beta^2 + 4\xi_t^2 \gamma^2)} \quad (9)$$

where $\bar{\beta}$ and $\bar{\xi}_t$ are the equivalent frequency ratio and damping ratio of the M-TMD, respectively.

Fig. 4 shows $\bar{\beta}$ and $\bar{\xi}_t$ varying with excitation frequency. In Den Hartog's book [36], if the natural frequency of the mass damper is always equal to the harmonic frequency of the external force and its damping ratio is zero, i.e., $\beta = \gamma$, $\xi_t = 0$, the structural vibration will be eliminated. Coincidentally, the M-TMD with the frequency-dependent frequency ratio is closer to the ideal case ($\bar{\beta} = \gamma$) than the TMD with a constant frequency ratio, shown in Fig. 4(a). The equivalent damping ratio of the M-TMD is smaller than that of the TMD when $\gamma \approx 1$, shown in Fig. 4(b), to make the M-TMD be more effective than the TMD near original resonance.

2.2. Maxwell tuned mass-damper-inerter with inerter-ground configuration

The inerter is a two-terminal device which provides a considerable force proportional to the relative acceleration between terminals [37,38],

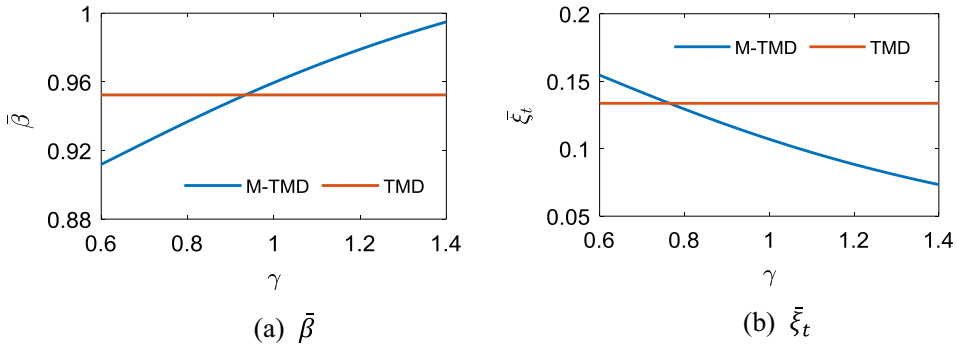


Fig. 4. The equivalent frequency ratio and damping ratio varying with excitation frequency.

$$F = m_e(\ddot{x}_2 - \ddot{x}_1) \quad (10)$$

where $\ddot{x}_1$ and $\ddot{x}_2$ are the absolute accelerations of two ends of an inerter, in Fig. 5, $\ddot{x}_1$ and $\ddot{x}_2$ are the zero and absolute acceleration of the auxiliary mass; m_e is the inertance or equivalent mass, with the unit of kilogram, a considerable m_e can be realized by the rack-and-pinion inerter, ball-screw inerter, and fluid inerter [37].

The inerter-ground M-TMDI refers to the M-TMD in conjunction with an inerter connected to the fixed ground, shown in Fig. 5. Since there is no inerter force F applied on the primary structure, the equation of motion of the primary structure equipped an inerter-ground M-TMDI is the same as Eq. (1), the equation of motion of the inerter-ground M-TMDI can be expressed as

$$(m_t + m_e)\ddot{x}_t(t) + u(t) = 0 \quad (11)$$

Obviously, the added inerter is equivalent to increasing the mass of the auxiliary mass from m_t to $(m_t + m_e)$, and does not change the nature and proportion of restoring forces. Thus, the optimum parameters of the inerter-ground M-TMDI can be obtained by using the optimization result of the M-TMD presented in Table 1,

$$\beta_{M-TMDI} = \frac{\bar{\omega}_t}{\bar{\omega}_p} = \sqrt{\frac{1}{1 + \mu + d} \left(1 - \sqrt{\frac{\mu + d}{1 + \mu + d}} \right)} \quad (12)$$

$$\kappa_{M-TMDI} = \frac{k_m}{k_t} = 2(\mu + d) + 2\sqrt{(\mu + d)(1 + \mu + d)} \quad (13)$$

$$\xi_{M-TMDI} = \frac{c_m}{2(m_t + m_e)\bar{\omega}_t} = \sqrt{\frac{1 + r}{r} \cdot \frac{-b - \sqrt{b^2 - ac}}{a}} \quad (14)$$

where $d = m_e/m_p$, $\bar{\omega}_t = \sqrt{k_t/(m_t + m_e)}$ and $r = \sqrt{(1 + \mu + d)/(\mu + d)}$. If the inerter force F is written in the frequency domain, $F = -\omega^2 m_e x_t$, the inerter can be considered as a negative stiffness element. The equivalent stiffness of the inerter-ground M-TMDI is written as

$$k_e = \frac{\omega^2 c_m^2 k_m}{k_m^2 + \omega^2 c_m^2} + k_t - \omega^2 m_e \quad (15)$$

We can find that the equivalent stiffness has the high-static-low-dynamic characteristic, which helps limit the static stretching. Besides, since the mass ratio is increased from μ to $(\mu + d)$, the stroke of the auxiliary mass will be reduced greatly when absorbing the same vibration energy.

3. Generalized Maxwell tuned mass-damper-inerter

3.1. Equation of motion

Since the inerter-ground configuration, shown in Fig. 5, is not easy to implement for bridges, the end of the inerter is assumed to be connected to any point of the bridge deck, shown in Fig. 6. The analysis of the M-TMDI for the VIV mitigation is based on the configuration shown in Fig. 6. The blue line in Fig. 6 means a mode shape of the bridge subjected to the VIV, not the bridge itself.

Assume that n M-TMDIs are installed on the bridge deck, the equation governing the vertical motion of the bridge deck at section x is written as

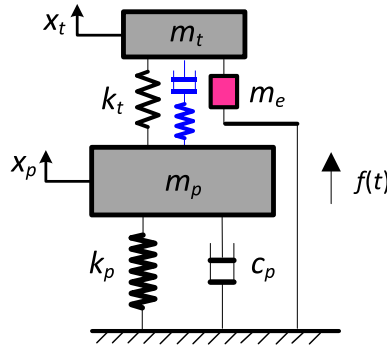


Fig. 5. A primary structure equipped with an inerter-ground M-TMDI.

$$m(x) \frac{\partial^2 y(x, t)}{\partial t^2} + c(x) \frac{\partial y(x, t)}{\partial t} + \frac{\partial^2 y(x, t)}{\partial x^2} \left[EI(x) \frac{\partial^2 y(x, t)}{\partial x^2} \right] = f_y(\dot{x}, x, t) + \bar{u}(x, t) \quad (16)$$

where y is the vortex-induced amplitude of the bridge deck; m , c , and EI are the mass per unit length, damping, and flexural rigidity in y -direction; f_y is the vortex-induced force acting on bridge deck; $\bar{u}$ is the M-TMDI force acting on the bridge deck. $\bar{u}$ is written as follows,

$$\bar{u}(x, t) = \sum_{i=1}^n \text{sign}(|x - x_{m,i}|) (k_{t,i} [y_{t,i}(t) - y(x, t)] + c_{m,i} [\dot{y}_{t,i}(t) - \dot{y}_{m,i}(t)]) + \text{sign}(|x - x_{b,i}|) m_{e,i} [\ddot{y}_{t,i}(t) - \ddot{y}(x, t)] \quad (17)$$

$$c_{m,i} [\dot{y}_{t,i}(t) - \dot{y}_{m,i}(t)] = \text{sign}(|x - x_{m,i}|) k_{m,i} [y_{m,i}(t) - y(x, t)] \quad (18)$$

where $y_{t,i}$ is the displacement of the i th M-TMDI; $y_{m,i}$ is the inner displacement of the Maxwell element in the i th M-TMDI; $x_{m,i}$ and $x_{b,i}$ are the position coordinates of the Maxwell element and inerter in the i th M-TMDI; $k_{t,i}$ is the stiffness of the i th M-TMDI; $k_{m,i}$ and $c_{m,i}$ are the stiffness and damping of the Maxwell element in the i th M-TMDI; $m_{e,i}$ is the equivalent mass provided by the inerter in the i th M-TMDI.

The equation of motion of the i th M-TMDI can be expressed as

$$m_{t,i} \ddot{y}_{t,i}(t) + k_{t,i} [y_{t,i}(t) - y_{x_{m,i}}(t)] + c_{m,i} [\dot{y}_{t,i}(t) - \dot{y}_{m,i}(t)] + m_{e,i} [\ddot{y}_{t,i}(t) - \ddot{y}_{x_{b,i}}(t)] = 0 \quad (19)$$

$$c_{m,i} [\dot{y}_{t,i}(t) - \dot{y}_{m,i}(t)] = k_{m,i} [y_{m,i}(t) - y_{x_{m,i}}(t)] \quad (20)$$

where $m_{t,i}$ is the mass of the i th auxiliary mass; $y_{x_{m,i}}$ and $y_{x_{b,i}}$ are the vortex-induced amplitudes of the bridge deck at sections $x_{m,i}$ and $x_{b,i}$.

Introduce $y(x, t) = \Phi(x) \mathbf{q}(t)$, and integrate from 0 to L and left multiply by $\mathbf{M}^{-1} \Phi^T$ at each side of Eq. (14), where L is the span length, then Eq. (14) can be written as

$$\ddot{\mathbf{q}} + \Lambda \dot{\mathbf{q}} + \Omega \mathbf{q} = \mathbf{F}(t) + \mathbf{U}(t) \quad (21)$$

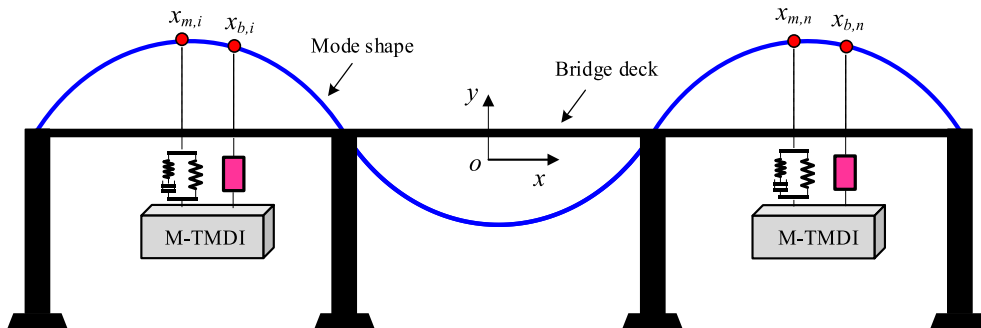


Fig. 6. The flexible bridge equipped with M-TMDIs.

$$\mathbf{U}(t) = \mathbf{M}^{-1} \left(\Phi^T \mathbf{T}_t \mathbf{k}_t \mathbf{y}_t + \Phi^T \mathbf{T}_t \mathbf{c}_m (\dot{\mathbf{y}}_t - \dot{\mathbf{y}}_m) + \Phi^T \mathbf{T}_b \mathbf{m}_e \ddot{\mathbf{y}}_t - \Phi^T \mathbf{T}_b \mathbf{m}_e \mathbf{T}_b^T \Phi \ddot{\mathbf{q}} - \Phi^T \mathbf{T}_t \mathbf{k}_t \mathbf{T}_t^T \Phi \mathbf{q} \right) \quad (22)$$

where $\Phi = [\varphi_1, \dots, \varphi_j, \dots, \varphi_N]$ is the mode shape matrix of the bridge deck; φ_j is the mode shape vector of the j th mode; $\mathbf{q} = \{q_1, \dots, q_j, \dots, q_N\}^T$ is the modal coordinate vector; q_j is the modal coordinate of the j th mode; N is the number of the mode; $\mathbf{y}_t = \{y_{t,1}, \dots, y_{t,i}, \dots, y_{t,n}\}^T$ is the displacement response vector of n M-TMDIs; $\mathbf{M} = \int_0^L \Phi^T(x) m(x) \Phi(x) dx$ is the modal mass matrix; $\Lambda = \text{diag}\{2\xi_1\omega_1, \dots, 2\xi_j\omega_j, \dots, 2\xi_N\omega_N\}$, $\Omega = \text{diag}\{\omega_1^2, \dots, \omega_j^2, \dots, \omega_N^2\}$; ω_j and ξ_j are the natural frequency and damping ratio for the j th mode; $\mathbf{T}_t$ and $\mathbf{T}_b$ are the position matrices of Maxwell elements and inerters; $\mathbf{F}$ is the vortex-induced force vector acting on bridge deck; $\mathbf{U}$ is the M-TMDI force vector acting on the bridge deck.

Similarly, the equation of motion for n M-TMDIs can be written as

$$\mathbf{m}_t \ddot{\mathbf{y}}_t + \mathbf{k}_t \mathbf{y}_t - \mathbf{k}_t \mathbf{T}_t^T \Phi \mathbf{q} + \mathbf{c}_m (\dot{\mathbf{y}}_t - \dot{\mathbf{y}}_m) + \mathbf{m}_e \ddot{\mathbf{y}}_t - \mathbf{m}_e \mathbf{T}_b^T \Phi \ddot{\mathbf{q}} = 0 \quad (23)$$

$$\mathbf{m}_t \ddot{\mathbf{y}}_t + \mathbf{k}_t \mathbf{y}_t - \mathbf{k}_t \mathbf{T}_t^T \Phi \mathbf{q} + \mathbf{k}_m \mathbf{y}_m - \mathbf{k}_m \mathbf{T}_t^T \Phi \mathbf{q} + \mathbf{m}_e \ddot{\mathbf{y}}_t - \mathbf{m}_e \mathbf{T}_b^T \Phi \ddot{\mathbf{q}} = 0 \quad (24)$$

where $\mathbf{m}_t = \text{diag}\{m_{t,1}, \dots, m_{t,i}, \dots, m_{t,n}\}$ is the mass matrix of the auxiliary mass; $\mathbf{m}_e = \text{diag}\{m_{e,1}, \dots, m_{e,i}, \dots, m_{e,n}\}$ is the mass matrix of the inerter; $\mathbf{k}_t = \text{diag}\{k_{t,1}, \dots, k_{t,i}, \dots, k_{t,n}\}$ is the stiffness matrix of the M-TMDI; $\mathbf{k}_m = \text{diag}\{k_{m,1}, \dots, k_{m,i}, \dots, k_{m,n}\}$ and $\mathbf{c}_m = \text{diag}\{c_{m,1}, \dots, c_{m,i}, \dots, c_{m,n}\}$ are the stiffness and damping matrices of the Maxwell element.

Combine Eqs. (21), (23) and (24), the equation of motion of the whole system is written as

$$\bar{\mathbf{M}} \ddot{\boldsymbol{\eta}} + \bar{\mathbf{C}} \dot{\boldsymbol{\eta}} + \bar{\mathbf{K}} \boldsymbol{\eta} = \bar{\mathbf{F}}(t) \quad (25)$$

where $\boldsymbol{\eta} = \begin{Bmatrix} \mathbf{q} \\ \mathbf{y}_t \\ \mathbf{y}_m \end{Bmatrix}$ is modal coordinate vector of the whole system; $\bar{\mathbf{M}} = \begin{bmatrix} \mathbf{I}_{N \times N} + \mathbf{M}^{-1} \Phi^T \mathbf{T}_b \mathbf{m}_e \mathbf{T}_b^T \Phi & -\mathbf{M}^{-1} \Phi^T \mathbf{T}_b \mathbf{m}_e & \mathbf{0}_{n \times n} \\ -\mathbf{m}_e \mathbf{T}_b^T \Phi & \mathbf{m}_t + \mathbf{m}_e & \mathbf{0}_{n \times n} \\ -\mathbf{m}_e \mathbf{T}_b^T \Phi & \mathbf{m}_t + \mathbf{m}_e & \mathbf{0}_{n \times n} \end{bmatrix}$,

$\bar{\mathbf{C}} = \begin{bmatrix} \Lambda & -\mathbf{M}^{-1} \Phi^T \mathbf{T}_t \mathbf{c}_m & \mathbf{M}^{-1} \Phi^T \mathbf{T}_t \mathbf{c}_m \\ \mathbf{0}_{N \times N} & \mathbf{c}_m & -\mathbf{c}_m \\ \mathbf{0}_{n \times N} & \mathbf{0}_{n \times n} & \mathbf{0}_{n \times n} \end{bmatrix}$, and $\bar{\mathbf{K}} = \begin{bmatrix} \Omega + \mathbf{M}^{-1} \Phi^T \mathbf{T}_t \mathbf{k}_t \mathbf{T}_t^T \Phi & -\mathbf{M}^{-1} \Phi^T \mathbf{T}_t \mathbf{k}_t & \mathbf{0}_{n \times n} \\ -\mathbf{k}_t \mathbf{T}_t^T \Phi & \mathbf{k}_t & \mathbf{0}_{n \times n} \\ -\mathbf{k}_t \mathbf{T}_t^T \Phi - \mathbf{k}_m \mathbf{T}_t^T \Phi & \mathbf{k}_t & \mathbf{k}_m \end{bmatrix}$ are the mass, damping,

and stiffness matrices of the whole system; $\bar{\mathbf{F}}(t) = \begin{Bmatrix} \mathbf{F} \\ \mathbf{0}_{n \times 1} \\ \mathbf{0}_{n \times 1} \end{Bmatrix}$ is the vortex-induced force vector of the whole system.

The semi-empirical nonlinear model [30–33] is widely used to describe the vortex-induced force acting on the bridge deck. The nonlinear model is written as

$$f_y(\dot{x}, x, t) = \rho U^2 D \left[Y_1(K) \left(1 - \varepsilon \frac{y^2}{D^2} \right) \frac{\dot{y}}{U} + Y_2(K) \frac{y}{D} + \frac{1}{2} C_L(K) \sin(\omega t + \theta) \right] \quad (26)$$

where ρ , U , and D are the air density, mean wind velocity, and feature size of the bridge section; Y_1 , Y_2 , and ε are the aerodynamic parameters obtained by wind tunnel tests; C_L and θ are the lift coefficient and initial phase; $K = \omega D/U$ is defined as reduced frequency; ω is the vortex-shedding frequency. The second item on the right of Eq. (26) can be dropped when the bridge is within the lock-in region [30–33], so that the generalized vortex-induced force for the j th mode can be written as

$$F_j(\dot{x}, x, t) = \rho U^2 D \int_0^L \left[Y_1 \left(\varphi_j^2(x) - \frac{\varepsilon}{D^2} \varphi_j^4(x) q_j^2 \right) \frac{\dot{q}_j}{U} + \frac{1}{2} C_L \sin(\omega t + \theta) \varphi_j(x) \right] dx \quad (27)$$

According to Eq. (27), the vortex-induced force vector in Eq. (21) can be expressed as

$$\mathbf{F} = \{F_1/M_1, F_2/M_2, \dots, F_N/M_N\}^T$$

3.2. Optimum design parameters

It is noted that Eqs. (12)–(14) cannot be directly adopted as optimization solutions of the M-TMDI, because the effect of the inerter location on the M-TMDI control is not considered. In this section, the effect of the inerter location will be removed by defining an effective mass ratio, so that Eqs. (12)–(14) can be adopted for bridges.

Before deriving the effective mass ratio, the system is simplified by using three assumptions or facts. First, only a vibration mode is considered in the M-TMDI control of the VIV in bridges, because wind tunnel test and filed test results showed that the VIV in bridges behaves a single-mode vibration, vertical-mode or torsional-mode [30–33]. Second, n M-TMDIs have identical parameters and locations. Third, the equivalent mass provided by inerters is much larger than the mass of the auxiliary mass, because the better static stretching and stroke reductions need greater inerter participation in the M-TMDI [24,25].

Based on the single-mode vibration and identical M-TMDI, shown in Fig. 7, Eqs. (21)–(24) can be simplified as

$$\ddot{q}_j + 2\xi_j\omega_j\dot{q}_j + \omega_j^2 q_j = F_j(t) + u_j(t) \quad (28)$$

$$u_j(t) = \frac{1}{M_j} \left[\varphi_j^T \mathbf{t}_t k_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + \varphi_j^T \mathbf{t}_t c_t (\dot{y}_t - \dot{y}_m) + \varphi_j^T \mathbf{t}_b m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) \right] \quad (29)$$

$$m_t \ddot{y}_t + k_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + c_m (\dot{y}_t - \dot{y}_m) + m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) = 0 \quad (30)$$

$$m_t \ddot{y}_t + k_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + k_m (y_m - \mathbf{t}_t^T \varphi_j q_j) + m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) = 0 \quad (31)$$

where $\mathbf{t}_t$ and $\mathbf{t}_b$ are position vectors of Maxwell elements and inerters. According to Eq. (6), the Maxwell element in parallel with a spring can be written as a Kelvin element with frequency dependence, so that the equivalent stiffness $\bar{k}_t$ and equivalent damping $\bar{c}_t$ are introduced into Eqs. (29)–(30) to remove the inner variables y_m and $\dot{y}_m$, where $\bar{k}_t = \omega^2 c_m^2 k_m / (k_m^2 + \omega^2 c_m^2) + k_t$ and $\bar{c}_t = c_m k_m^2 / (k_m^2 + \omega^2 c_m^2)$. Then, Eqs. (29)–(31) can be rewritten as,

$$u_j(t) = \frac{1}{M_j} \left[\varphi_j^T \mathbf{t}_t \bar{k}_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + \varphi_j^T \mathbf{t}_t \bar{c}_t (\dot{y}_t - \mathbf{t}_t^T \varphi_j \dot{q}_j) + \varphi_j^T \mathbf{t}_b m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) \right] \quad (32)$$

$$m_t \ddot{y}_t + \bar{k}_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + \bar{c}_t (\dot{y}_t - \mathbf{t}_t^T \varphi_j \dot{q}_j) + m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) = 0 \quad (33)$$

Based on the great inerter participation, $m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j)$ will be much larger than $m_t \ddot{y}_t$ in Eq. (33), i.e., the inerter dominates the optimization solutions of the M-TMDI. Ignoring $m_t \ddot{y}_t$, Eq. (33) is written as

$$\bar{k}_t (y_t - \mathbf{t}_t^T \varphi_j q_j) + \bar{c}_t (\dot{y}_t - \mathbf{t}_t^T \varphi_j \dot{q}_j) + m_e (\ddot{y}_t - \mathbf{t}_b^T \varphi_j \ddot{q}_j) = 0 \quad (34)$$

Furthermore, introducing new variables $z = y_t - \mathbf{t}_b^T \varphi_j q_j$ and $v = \mathbf{t}_t^T \varphi_j - \mathbf{t}_b^T \varphi_j$, Eqs. (32) and (34) can be rewritten as

$$u_j(t) = \frac{1}{M_j} \left[\varphi_j^T \mathbf{t}_t \bar{k}_t (z - v q_j) + \varphi_j^T \mathbf{t}_t \bar{c}_t (\dot{z} - v \dot{q}_j) + \varphi_j^T \mathbf{t}_b m_e \ddot{z} \right] \quad (35)$$

$$m_e \ddot{z} + \bar{k}_t (z - v q_j) + \bar{c}_t (\dot{z} - v \dot{q}_j) = 0 \quad (36)$$

According to Eq. (36), the relation between z and q_j in the frequency domain can be given by

$$z = \frac{(i\omega \bar{c}_t + k_t)}{(-\omega^2 m_e + i\omega \bar{c}_t + \bar{k}_t)} v q_j \quad (37)$$

Taking Fourier transform of Eq. (35), and substituting Eq. (37) into Eq. (35), Eq. (35) can be written as,

$$u_j(\omega) = \frac{(\varphi_j^T \mathbf{t}_b - \varphi_j^T \mathbf{t}_t) m_e (-v)}{M_j} \frac{\omega^2 (i\omega \bar{c}_t + \bar{k}_t)}{(-\omega^2 m_e + i\omega \bar{c}_t + \bar{k}_t)} q_j \quad (38)$$

It is noted that Eq. (38) is the same as the expression of a single-DOF structure equipped with a TMD [5,36], with an effective mass ratio $(\varphi_j^T \mathbf{t}_b - \varphi_j^T \mathbf{t}_t) m_e (\varphi_j^T \mathbf{t}_b - \varphi_j^T \mathbf{t}_t) / M_j$.

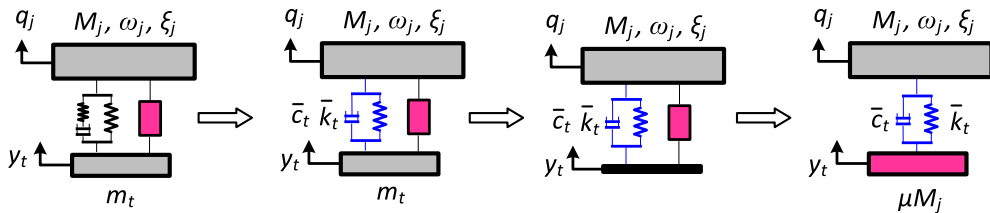


Fig. 7. The analytical model of damping system in the j th mode.

If the physical mass is considered, the effective mass ratio can be expressed as follows

$$\mu_j = \frac{(\varphi_j^T \mathbf{t}_b - \varphi_j^T \mathbf{t}_t) m_e (\varphi_j^T \mathbf{t}_b - \varphi_j^T \mathbf{t}_t) + \varphi_j^T \mathbf{t}_t m_t \varphi_j^T \mathbf{t}_t}{M_j} \quad (39)$$

At last, the optimization solutions of β_j , κ_j , and ξ_{tj} can be obtained by using μ_j instead of $(\mu + d)$ in Eqs. (12)–(14). Furthermore, the optimal parameters of the M-TMDI can be expressed as follows

$$k_t = (m_e + m_t) \beta_j^2 \omega_j^2 \quad (40)$$

$$k_m = \kappa_j k_t \quad (41)$$

$$c_m = 2(m_e + m_t) \beta_j \omega_j \xi_{tj} \quad (42)$$

In order to verify the optimization solutions, Fig. 8 shows the control result of the optimally design M-TMDI. In the M-TMDI control, two cases are considered, $m_e/M_j = 1$ and $m_e/M_j = 10\%$, other parameters are the same, $m_t/M_j = 0.5\%$, $\varphi_j^T \mathbf{t}_b = 0.5$, $\varphi_j^T \mathbf{t}_t = 1$. In the M-TMD control, $m_t/M_j = 25.5\%$ and $m_t/M_j = 3\%$ to ensure the same effective mass ratio. The solid and dashed lines refer to the control results using the M-TMDI and M-TMD, respectively. Like other optimally designed mass dampers, the two equal peaks of the displacement DAF curve are observed in the M-TMDI control, indicating that the M-TMDI parameters are optimal. A good match between the two types of displacement DAF curves indicates that the optimally designed M-TMDI retains the advantage of the optimally designed M-TMD.

4. Numerical example

The continuous steel box-girder bridge used in the numerical example is a 6×110 m standard link of the Hong Kong-Zhuhai-Macao Bridge deep-water non-navigable bridge, with a height of 4.5 m and width of 33.1 m section, shown in Fig. 9. The finite element model is established using Ansys software, the information on the finite element model can be found in [39]. The finite element model accuracy was examined by the free vibration test of a 1:70-scaled full-bridge aeroelastic model [39,40]. The finite element analysis shows that the generalized mass and natural frequency for the fundamental vertical mode of the bridge are 8888.9 ton and 0.806 Hz. The damping ratio for the fundamental vertical mode of the bridge is assumed as 0.3%. Fig. 10 shows the fundamental vertical mode shape of the bridge.

Wind tunnel tests of the 1:50 scale section model, 1:20 scale section model, and 1:70 scale full-bridge aeroelastic model were conducted to evaluate the VIV of the bridge [40]. The results showed that a vortex-induced resonance as the fundamental vertical mode occurs when the wind velocity is 27–35 m/s. The maximum vortex-induced amplitude is 140 mm which exceeds the limit suggested by China code, 50 mm. The aerodynamic parameters were identified using a simplified decay-to-resonance method [41], $Y_1 = 15.674$, $\varepsilon = 2.9 \times 10^3$. Based on the data from the finite element analysis and wind tunnel tests, the vortex-induced resonance of the bridge and the M-TMDI control can be simulated in Matlab by solving Eq. (25). The Newmark- β method in conjunction with the direct iteration method [42] is used as the numerical method in simulation. The initial amplitude of the bridge is assumed as 100 mm. Fig. 11 shows the simulated vortex-induced resonance of the bridge. It can be seen that, in the lock-in region, the VIV of the bridge exhibits a limit cycle oscillation with a steady-state amplitude of 140 mm, shown in Fig. 11(a), which reproduces the vortex-induced resonance in wind tunnel tests. If the span-wise correlation effect is considered, the steady-state amplitude of the bridge can research 161.7 mm, shown in Fig. 11(b). Andersen et al. [31] suggested that a small lift coefficient term C_L should be included in the lock-in region, as a result, the steady-state amplitude can research 173.4 mm by adjusting the lift coefficient term reasonably, shown in Fig. 11(c).

In the real bridge, the eight TMDs with a total mass of 3.2×10^4 kg and 300 mm stroke, were installed in the center of the 2nd and 5th spans to mitigate the VIV. In the simulation, the M-TMDIs are used to mitigate the VIV of the 6×110 m bridge,

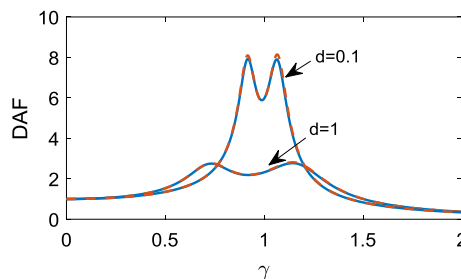


Fig. 8. The control result of the optimally design M-TMDI.

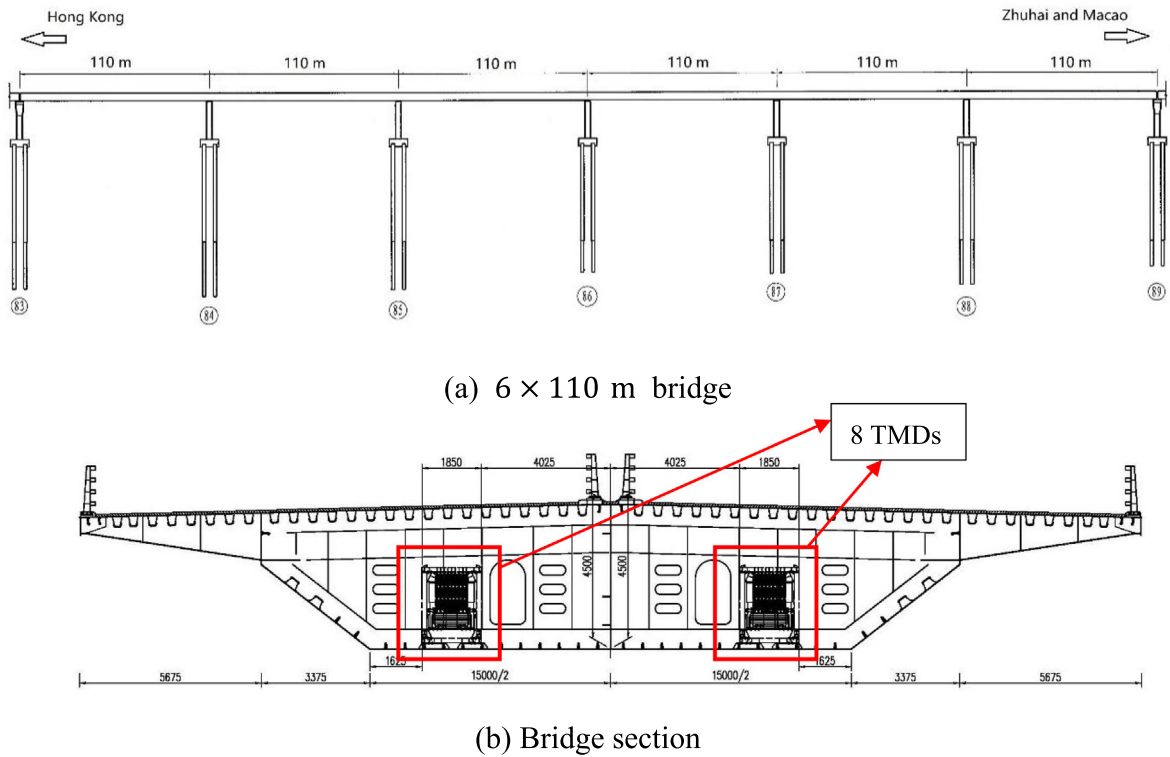


Fig. 9. The continuous steel box-girder bridge.

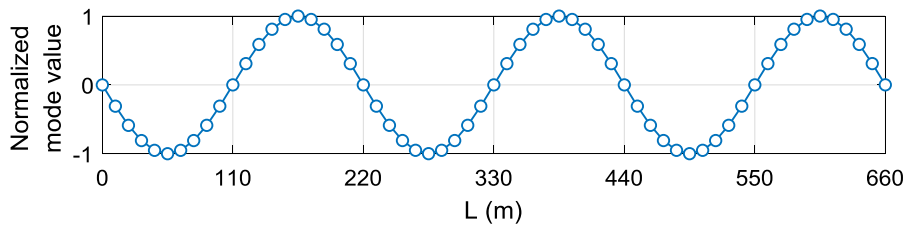


Fig. 10. The fundamental vertical mode shape of the bridge.

the TMD and M-TMD are used as comparisons. According to Den Hartog's formulas [36], the frequency ratio and damping ratio of the TMD are 0.996 and 3.67, respectively. According to Asami's formulas [27], the frequency ratio, stiffness ratio and damping ratio of the M-TMD are 0.968, 0.127, and 5.9%. For the M-TMDI, the inertial parameter is designed as $d=0.2$, and the physical mass is 3.2×10^4 kg. The Maxwell element is installed in the position that is 6 m from the pier, and two ends of the inerter are connected to the pier and auxiliary mass, respectively, i.e., $\mathbf{t}_t^T \boldsymbol{\varphi}_j = 0.16$, $\mathbf{t}_b^T \boldsymbol{\varphi}_j = 0$. The optimal parameters of the M-TMDI are determined according to Eqs. (40)–(42). The M-TMDI can adopt the lever-type TMD scheme [43] as the potential implementation for bridge structures.

The result of the three displacement DAF curves with two equal peaks indicates that the design parameters of the TMD, M-TMD, and M-TMDI are optimal, and the M-TMDI has the best control effect, shown in Fig. 12. Fig. 13 shows the static stretching of stiffness elements due to gravity. It can be seen that the TMD and M-TMD have large static stretchings, which requires more installation consideration for bridge structures. However, the static stretching of stiffness elements in the M-TMDI is small enough due to the added inerters, so that the static stretching cannot be a design concern.

Fig. 14 shows the control results of the three types of mass dampers. It can be seen that the limit cycle oscillation ceases to exist within 0 s to 60 s, which indicates that the vortex-induced resonance in the bridge is well mitigated by using the TMD, M-TMD and M-TMDI. Furthermore, the amplitude of the bridge is reduced to the limit of 50 mm after about 8.5 cycles, 7.5 cycles, and 6 cycles using the TMD, M-TMD, and M-TMDI, respectively, which indicates that the M-TMDI has the best control

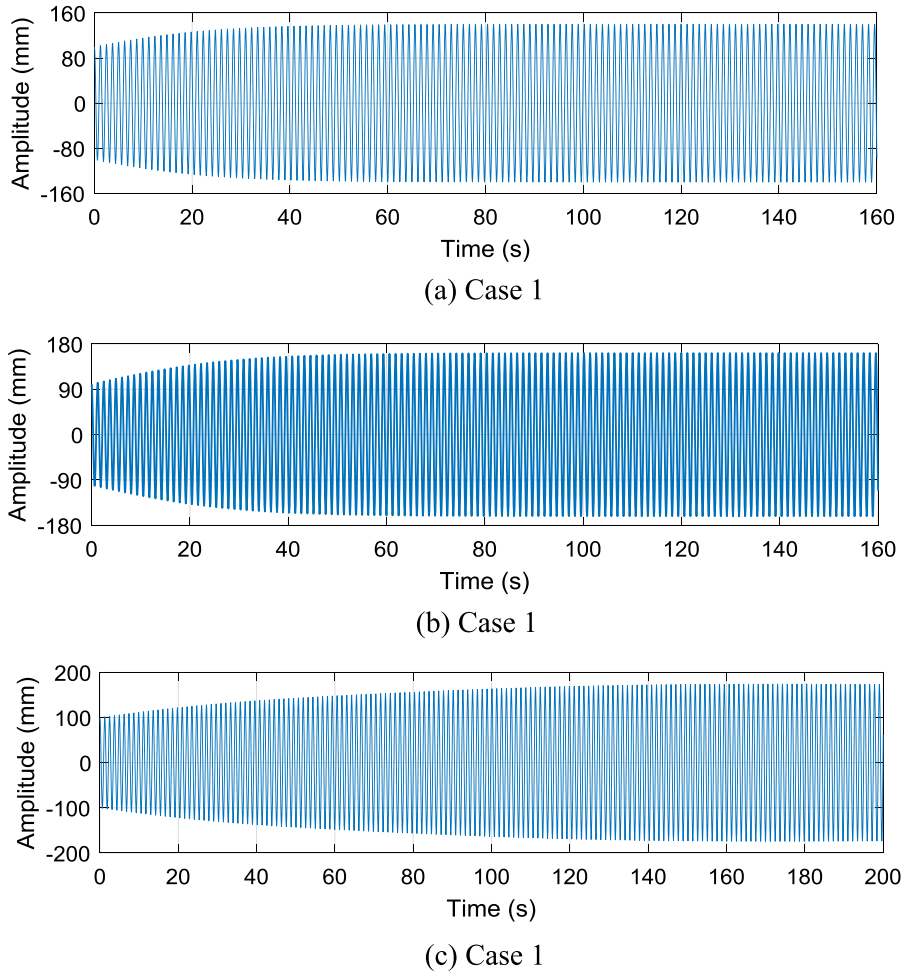


Fig. 11. The vortex-induced amplitude of the bridge.

effect for the VIV. Since the bridge-damper system with the closely spaced frequency has a serious couple effect, the transient response of the controlled bridge will exhibit the beat phenomenon decay instead of the exponential decay [44]. Thus, the amplitude of the bridge has a process of first increasing and then decaying within 15 s to 40 s, while the M-TMDI still has the best control effect. Within 60 s to 120 s, the steady-state amplitudes of the bridge induced by the lift coefficient term are 11.39 mm, 8.70 mm, 6.87 mm using the TMD, M-TMD and M-TMDI, respectively, which indicates that the M-TMDI is the most effective one. The improvement ratios of the M-TMD control and M-TMDI control are 23.61% and 39.68% compared to the TMD control, which are basically consistent with the result of the displacement DAF curve, 18.33% and 37.65%.

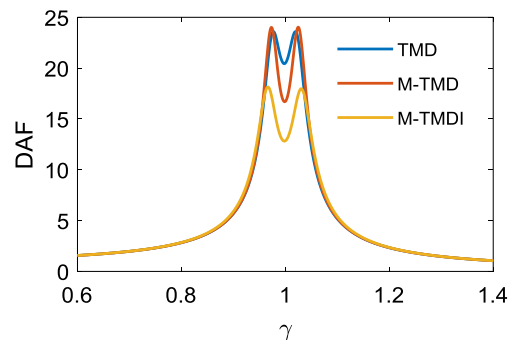


Fig. 12. The displacement DAF curves using the three mass dampers.

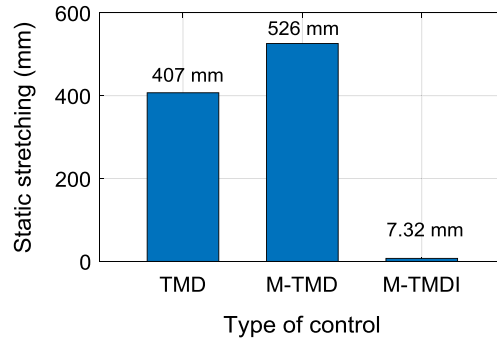


Fig. 13. The static stretching of stiffness elements.

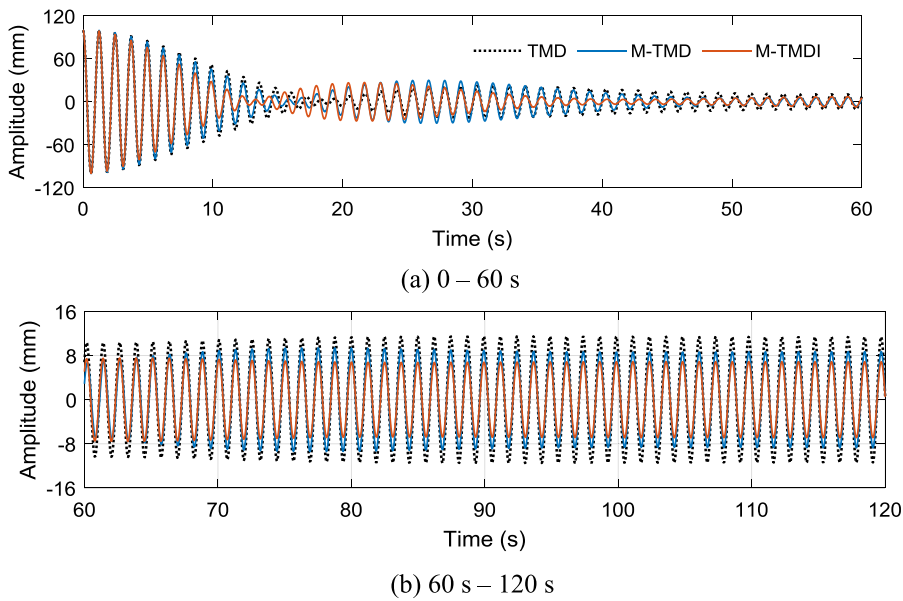


Fig. 14. The control results of the three types of mass dampers.

Fig. 15 shows the mass strokes of the three types of mass dampers. Within 0 s to 60 s, the maximum mass strokes are 879 mm, 979 mm and 122.7 mm using the TMD, M-TMD and M-TMDI, respectively. Within 60 s to 120 s, the maximum mass strokes are 156.3 mm, 148.2 mm and 16.51 mm using the TMD, M-TMD and M-TMDI, respectively. Those values are obtained based on the assumed initial displacement, so that they cannot be a basis for the stroke design but can illustrate the advantage of the M-TMDI in the stroke limitation. The obvious advantage of the M-TMDI is due to the large equivalent mass provided by inerters, which share a lot of kinetic energy from the physical mass.

To further illustrate the control effect of the M-TMDI, the other two M-TMDIs named as P-2 and P-3 are designed. P-1 refers to the M-TMDI mentioned above. P-2 refers to that the Maxwell element is installed 11 m from the pier, i.e., $\mathbf{t}_t^T \boldsymbol{\varphi}_j = 0.3088$, and other parameters are the same as those of P-1. P-3 refers to that the inertial parameter is designed as $d = 0.3$, and other parameters are the same as those of P-1. Fig. 16 shows the control result of these three M-TMDIs. It can be seen that the transient amplitude of the bridge is reduced to 50 mm after only 3 cycles, 5 cycles under P-2 and P-3, which indicates that the two M-TMDIs have better control effect than P-1 in the VIV mitigation. In addition, the better control effect is also found in the mitigation of the steady-state amplitude, shown in Fig. 16(b). This is because that the effective mass ratio is improved a lot in the P-2 and P-3 by increasing the value of $(\mathbf{t}_t^T \boldsymbol{\varphi}_j - \mathbf{t}_b^T \boldsymbol{\varphi}_j)$ and m_e , respectively. In practice, we can adopt the two design strategies in combination to improve the control performance of the M-TMDI considering the control cost and installation difficulty.

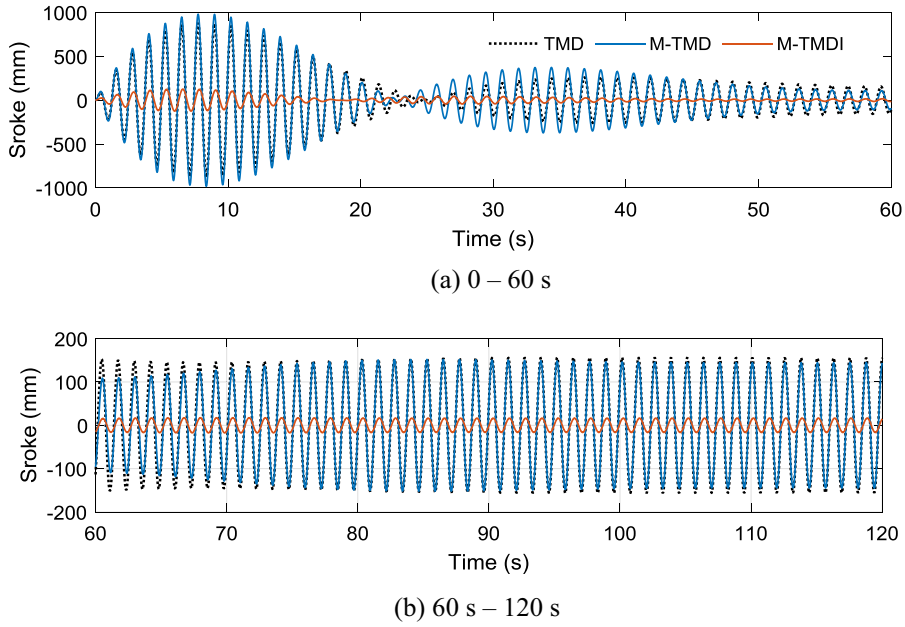


Fig. 15. The mass strokes of the three types of mass dampers.

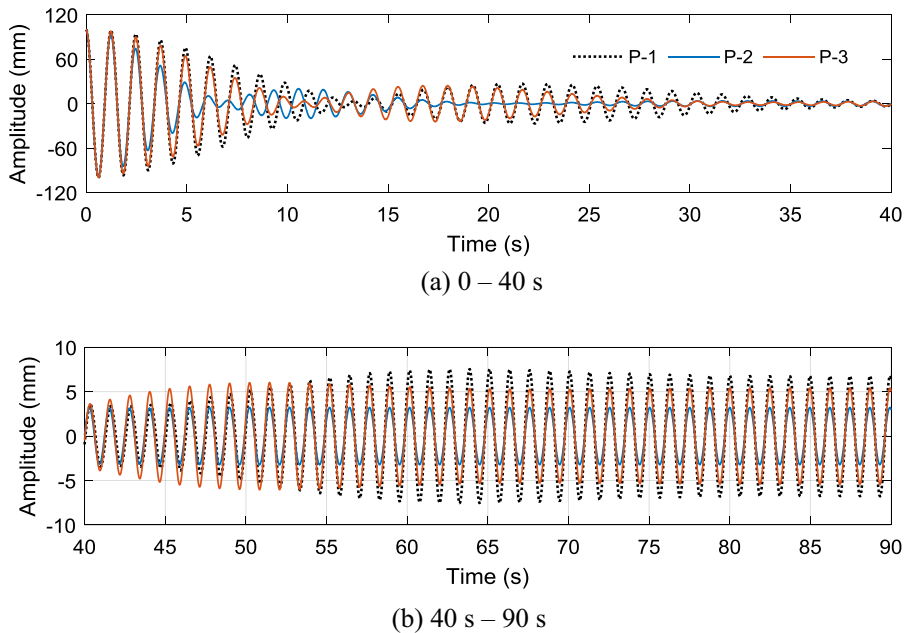


Fig. 16. The control result of these three M-TMDIs.

5. Conclusions

This paper proposes a passive TMD named the M-TMDI, which consists of a three-element TMD and an inerter. The M-TMDI has advantages of the three-element TMD and TMDI, including a good control effect in the resonance, small static stretching and small stroke. Based on these features, the M-TMDI is used to mitigate the VIV in bridges. The closed-form optimization solutions for the M-TMDI with any inerter location are developed. A numerical example of a continuous steel box-girder subjected to the VIV is performed to confirm the optimal design and superiority of the M-TMDI control. Based on the study, some conclusions can be obtained,

- (1) In the two-DOF system under harmonic force excitation, the optimization solutions of the M-TMD can be used to calculate the optimal parameters of the M-TMDI by changing the effective mass ratio. In a multiple-DOF bridge equipped with the M-TMDI, the proposed design formulas can consider the effect of the inerter location on the optimal parameters. The optimal control effect of the M-TMDI greatly depends on the defined effective mass ratio, which is function of the inerter location, mode shape, physical mass, and inertial mass.
- (2) Among three mass dampers, the M-TMDI is the most effective in the VIV mitigation of bridges when they have the same physical mass, both in the transient amplitude mitigation and steady-state amplitude mitigation. The mass stroke and static stretching of the M-TMDI is much less than those of the TMD and the M-TMD.

Data availability statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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